Purchase-to-pay process mining: where the process breaks

Tina Liang | Independent case study on the public BPI Challenge 2019 log (1.6M events, 251,652 purchase order items after cleaning) | github.com/tinabigbig

The question

Where does a large company's purchasing process break, how big is each problem, and what should finance and procurement look at first?

What I found

Rework. 16.9% of items in the main category (37,261 of 220,948) had a price or quantity change, deletion or cancellation. Trading & End Products is reworked at 32.5%, almost three times the rate in Sales.

Payment control. The strict invoice-after-goods control held. Judged at the moment each invoice was cleared, 346 of 183,292 paid 3-way match items (0.19%, EUR 1.97M at order value) had no valid goods receipt: none yet (89), none at all (176), or all cancelled (81). These are review opportunities, not confirmed breaches. Another 390 look like incomplete history.

Standard process. Replaying every main-category item against a model of the 10 most common paths (PM4Py), 82.7% fit (182,794 of 220,948). The misfits are mostly corrections: cancelled invoices, debit memos, quantity and price changes.

Speed. Median 77 days from order item to invoice clearing. 27.5% of paid items had a payment block removed; they took a median 85.5 days against 72.2 (an association, not proof of cause).

What it might be worth

113,248 manual touches (rework events and payment block removals) come from the data. Minutes, cost and the avoidable share are assumptions, so the value is a range.

Scenario	Min/touch	EUR/hour	Avoidable	Avoidable cost
Low	5	45	25%	EUR 106,170
Base	10	60	40%	EUR 452,992
High	15	75	55%	EUR 1,167,870

What I would do next

1. Review the 346 flagged payments, largest first; confirm the 390 incomplete-history items. 2. Start rework reduction in Trading & End Products: ask where quantity and price changes come from. 3. Find out why payment blocks are set. 4. Time a sample of real touches and rerun the value range.

How it was done

Pandas for the counts, PM4Py for the process map and conformance check, 12 tests; headline numbers recomputed with separate code on a hand-built log. One row is one event; duplicates dropped; 82 items with impossible dates removed before counting. Limits: anonymized data shows what happened, not why; 'Clear Invoice' is used as the payment step.

Data: van Dongen, B.F. (2019) BPI Challenge 2019, 4TU.ResearchData, CC BY 4.0, doi 10.4121/uuid:d06aff4b-79f0-45e6-8ec8-e19730c248f1